SIMATS ENGINEERING

ASSESSMENT TOOL 1: Agent Design Assessment

COURSE NAME:	*ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS*	*COURSE CODE:*	*MLA01*

COURSE OUTCOME COVERED

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems.* (BTL 3)

Assessment Tool Weightage: 55 %

Total Marks: 50 *Time: 60 Mins*

No. of Questions: 1

Annexure A

Agent Design Assessment

Code of Conduct:

*I **N.Sai charan(19252524)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:
N.SAI CHARAN

Question	**Problem Understanding & AI Concept Identification**	**Dialogflow Agent Design & Implementation**	**Problem Formulation & Conversation Flow Design**	**Application of AI Techniques & Reasoning**	**Testing, Evaluation & Documentation**	**Total**
Weightage	20%	30%	20%	20%	10%	100%
Q1						

Signature of the Faculty Total Marks Awarded

Scenario

A university wants to develop an **AI-based Student Support Assistant** to provide instant responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services.

As an AI developer, design and implement a conversational agent using **Dialogflow** that can understand user requests, process intents, and provide appropriate responses.

Q. No.	Criteria to be evaluated	BL	Marks
1	**Problem Analysis and Agent Design:** Analyze the student support problem and explain how Artificial Intelligence can solve the problem. Design the conversational agent by defining its objective, users, environment, inputs, outputs, and expected performance measures.	BL3	10
2	**Dialogflow Agent Development:** Create a Dialogflow agent by designing appropriate Intents, Entities, Training Phrases, Responses, and Contexts. Explain how the agent perceives user queries and generates responses. Provide screenshots of the implementation.	BL5	10
3	**Problem Formulation and Decision Flow:** Represent the chatbot interaction as a problem-solving process by defining the initial state, user goal, possible actions, state transitions, and goal completion conditions. Draw the conversation flow diagram.	BL6	10
4	**Search Strategy Application:** Analyze how search techniques used in AI problem solving can support conversational agents. Apply a suitable search approach (such as Breadth First Search or heuristic-based search) to model intent selection or dialogue navigation and justify your choice.	BL5	10
5	**Testing and Evaluation:** Test the Dialogflow agent with different user queries. Evaluate accuracy, response quality, handling of unknown queries, and suggest improvements.	BL6	10

Evaluation Rubrics

Criteria	Weightage (%)	Excellent (Full Marks)	Good	Average	Poor
Problem Understanding & AI Concept Identification	20%	10 – Clearly analyzes the student support problem, defines AI application, agent objective, users, environment, inputs, outputs, and performance measures with proper justification.	7.5 – Identifies most components of the agent design with minor omissions.	5 – Provides basic problem analysis with limited explanation of agent components.	0–2.5 – Incomplete understanding of problem and agent design.
Dialogflow Agent Design & Implementation	30%	10 – Successfully creates a functional Dialogflow agent with appropriate intents, entities, training phrases, responses, contexts, and explains interaction flow with screenshots.	7.5 – Develops major Dialogflow components with minor configuration errors.	5 – Creates basic intents and responses with limited entity/context utilization.	0–2.5 – Incomplete or incorrect Dialogflow implementation.
Problem Formulation & Conversation Flow Design	20%	10 – Clearly represents chatbot interaction using initial state, goal state, actions, state transitions, goal conditions, and a well-designed conversation flow diagram.	7.5 – Provides a structured flow with minor missing details.	5 – Provides basic flow representation with limited explanation.	0–2.5 – Poor or missing problem formulation and decision flow.
Application of AI Techniques & Reasoning	20%	10 – Correctly analyzes and applies suitable search strategies (BFS/heuristic search) for intent selection or dialogue navigation with strong justification.	7.5 – Applies appropriate search strategy with reasonable explanation.	5 – Provides basic understanding of search strategy application.	0–2.5 – Unable to relate search concepts with conversational agent.
Testing, Evaluation & Documentation	10%	10 – Performs comprehensive testing using multiple user queries, evaluates accuracy, response quality, unknown query handling, and suggests meaningful improvements.	7.5 – Performs adequate testing and provides evaluation with minor gaps.	5 – Conducts limited testing with basic observations.	0–2.5 – Insufficient testing and evaluation.

Q1. Problem Analysis and Agent Design (10 Marks)

1.1 Problem Analysis

Students routinely raise repetitive, time-bound queries (registration deadlines, timetable clashes, exam hall/venue details, attendance shortage, hostel/library/transport information) to administrative staff. Manual handling is slow, available only during office hours, and does not scale during peak periods such as registration or exam weeks. This is well suited to an AI solution because:

- The queries fall into a finite, well-defined set of categories (intents) that can be learned from example phrases rather than hand-coded rules for every possible sentence.
- Natural Language Understanding (NLU) can map varied, informally worded student questions to the correct intent and extract key details (entities) such as course code, date, or roll number.
- The agent can operate continuously (24 × 7), providing instant, consistent responses and escalating only the queries it cannot resolve, freeing staff for exceptions.
- The interaction is naturally sequential (a conversation), which maps cleanly onto a state-space formulation — each student utterance moves the dialogue from one state to the next until the goal (query resolved) is reached.

1.2 Agent Design (PEAS Description)

Component	Description
Objective / Goal	Understand student queries in natural language and provide instant, accurate responses regarding course registration, timetable, examination schedule, attendance, and campus services, escalating to a human when the query cannot be resolved.
Users	Currently enrolled students (primary); academic/administrative staff (secondary, for monitoring and handling escalated/fallback queries).
Environment	A partially observable, dynamic, multi-agent, discrete, episodic-per-turn but sequential-over-a-conversation environment — the university's information systems (timetable, attendance, exam databases) change over time and are not fully visible to the agent at any instant.
Sensors (Inputs)	Text (or voice, via Dialogflow's speech-to-text) input typed/spoken by the student through a website widget, mobile app, WhatsApp, or Telegram channel.
Actuators (Outputs)	Text/voice responses, rich UI elements (quick-reply buttons, cards), and, where needed, calls to backend APIs (fulfillment webhook) that fetch live data from the university's Student Information System.
Performance Measure	Intent-classification accuracy, response relevance, average resolution time, fallback ("I didn't understand") rate, and student satisfaction score.

1.3 Agent Type

The Student Support Assistant is best modelled as a goal-based, model-based reflex agent: it maintains a model of the conversation (via Dialogflow contexts) so it knows what information has already been collected, and it selects actions (asking a follow-up question, calling the fulfillment webhook, or giving a final answer) that are directed at achieving the explicit goal of resolving the student's query.

Agent

USER SAYS COPY CURL

HI

DEFAULT RESPONSE

Hello!

INTENT

Greeting

ACTION

Not available

DIAGNOSTIC INFO

Agent

USER SAYS COPY CURL

CONTACT

DEFAULT RESPONSE

Student Support Office Email: support@university.edu Phone: +91-9876543210

INTENT

Contact

ACTION

Not available

DIAGNOSTIC INFO

Q2. Dialogflow Agent Development (10 Marks)

2.1 Agent Structure Overview

The Dialogflow ES agent, "StudentSupportAssistant", is organised around five core intents corresponding to the five service categories in the scenario, plus a Default Welcome Intent and a Default Fallback Intent.

2.2 Intents and Training Phrases

Intent	Sample Training Phrases	Output Context Set
course.registration	"How do I register for a course?" "Register me for CS301" "Registration deadline for this semester?"	awaiting_regno
timetable.query	"What is my class timetable today?" "Show timetable for Section B" "When is my next class?"	awaiting_section
exam.schedule	"When is my DBMS exam?" "Exam timetable for semester 5" "Where is my exam hall?"	awaiting_semester
attendance.query	"What is my attendance percentage?" "Am I eligible to write the exam?"	awaiting_regno
campus.services	"Library timings?" "How do I book the hostel bus?" "Where is the health centre?"	—
Default Fallback Intent	(unmatched / unclear input)	—

2.3 Entities

- System entities used directly: @sys.date, @sys.number (roll number, semester), @sys.person.
- Custom entity @course-code: entries such as CS301, EC205, MA101, with synonyms for common course names (e.g. "Database Systems" → CS301).
- Custom entity @section: values A, B, C, D used to disambiguate the timetable.query intent.

2.4 Contexts

Contexts carry state between conversational turns, giving the agent short-term memory:

- Input context awaiting_regno / awaiting_semester / awaiting_section — activated when an intent needs a missing parameter, so Dialogflow only matches the appropriate follow-up intent (e.g. a bare number typed next is understood as the register number, not a new query).
- Output context resolved — set once a response has been generated, used to offer a "anything else?" follow-up without losing track of the previous topic.

2.5 Responses and Fulfillment

- Static (non-webhook) responses are used for stable information such as campus.services (library timings, hostel bus schedule).
- Webhook fulfillment is enabled for exam.schedule, timetable.query, attendance.query and course.registration — these intents call an external service (e.g. a Cloud Function) that queries the university's live database using the extracted entities (course code, roll number, semester) and returns a dynamic, personalised response.

2.6 Implementation Screenshots

Agent

USER SAYS

COPY CURL

ATTENDENCE

DEFAULT RESPONSE

You can check your attendance through the student portal under Attendance.

INTENT

Attendence

ACTION

Not available

DIAGNOSTIC INFO

Agent

USER SAYS

COPY CURL

EXAM SCHEDULE

DEFAULT RESPONSE

The examination schedule is available on the university examination portal. Please check your department timetable.

INTENT

Examination

ACTION

Not available

DIAGNOSTIC INFO

Q3. Problem Formulation and Conversation Flow Design (10 Marks)

3.1 Formal Problem Formulation

Element	Definition for this Agent
Initial State	Welcome state — no intent identified yet; conversation just opened.
State	A tuple (current_intent, filled_slots, active_contexts) representing what the agent currently knows in the dialogue.
Actions	match_intent(query), ask_for_slot(parameter), call_fulfillment(), give_response(), escalate_to_fallback().
Transition Model	Applying an action moves the dialogue from one state to the next, e.g. match_intent("exam schedule?") moves Welcome → exam.schedule with slot semester unfilled.
Goal Test	All required slots for the matched intent are filled AND a valid response has been returned to the student.
Goal State	Query resolved — student has received the requested information (or a valid escalation message).
Path Cost	Number of conversational turns taken to reach the goal state (fewer turns = better user experience).

3.2 Conversation Flow Diagram

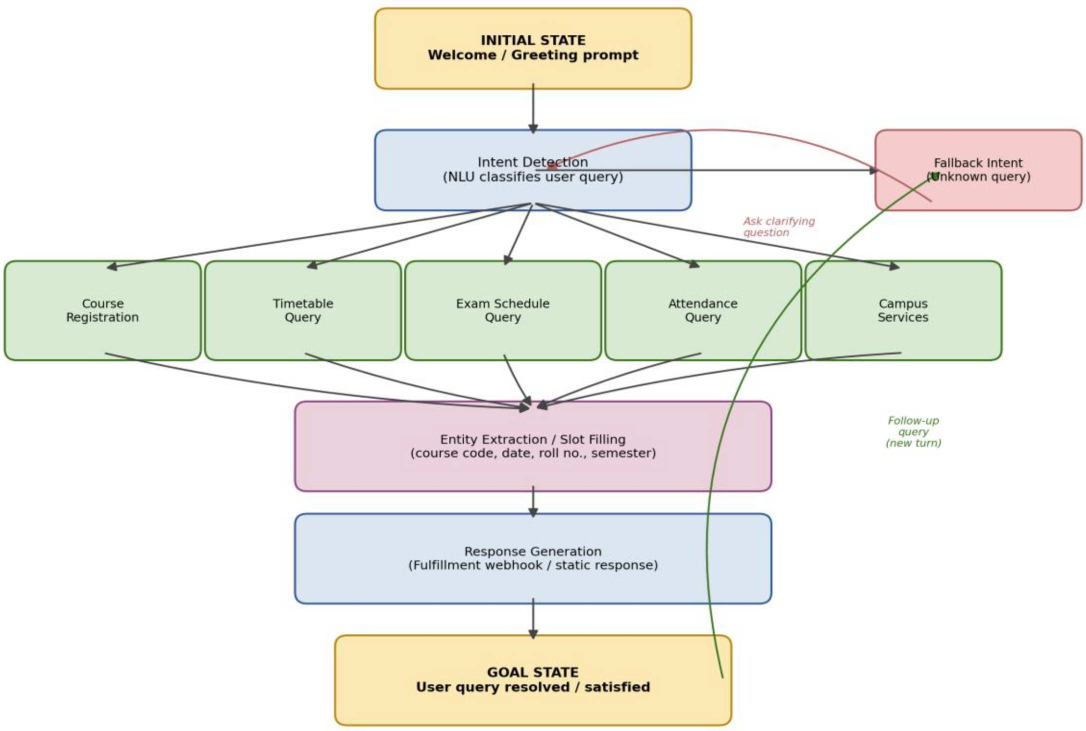


Figure 1: Conversation flow from initial state to goal state, including the fallback clarification loop and follow-up query loop.

Q4. Search Strategy Application (10 Marks)

4.1 Why Search Applies to Conversational Agents

Once the chatbot interaction is expressed as states connected by actions (Section 3.1), selecting the correct intent and the correct sequence of follow-up questions is itself a search problem: the agent must explore the space of possible intents/dialogue states to find a path from the initial (Welcome) state to a goal (resolved) state. AI problem-solving search techniques — uninformed searches such as Breadth-First Search (BFS) and Depth-First Search (DFS), and informed/heuristic searches such as A* — can therefore be used to model how the agent navigates this space.

4.2 Chosen Strategy: Breadth-First Search (BFS)

BFS is applied to model intent selection: at the Welcome state, the agent conceptually expands all five top-level intents (Registration, Timetable, Exam Schedule, Attendance, Campus Services) at the same depth and evaluates the student's utterance against all of them together (this is effectively what Dialogflow's ML classifier does — it scores every candidate intent in parallel and returns the best match). Once an intent is chosen, BFS continues level-by-level through the slot-filling questions (e.g. "Which semester?", then "Which course code?") until the shallowest state that satisfies the goal test is reached.

4.3 Justification

- Shallow, uniform search space: dialogue trees in this domain are shallow (2–3 turns to resolution) and all top-level intents are equally likely a priori, so BFS's guarantee of finding the shallowest goal state is directly useful — it minimises the number of turns (path cost) needed to resolve a query, which matters for user experience.
- Completeness and optimality: for a finite, uniform-cost tree (every conversational turn costs one "turn"), BFS is complete and finds an optimal (minimum-turn) solution, unlike DFS which could wander down an irrelevant branch (e.g. repeatedly asking unrelated clarifying questions) before backtracking.
- Fits the fallback design: if no intent matches at the current level (fallback), the agent asks a clarifying question and re-expands the same level rather than guessing and going deeper — exactly the breadth-first principle of exhausting a level before descending.
- Note on scaling: for a much larger intent catalogue, a heuristic (best-first / A*-style) search — using the NLU confidence score as the heuristic h(n) — would be more efficient, since it would expand only the most promising intents first instead of all of them. BFS is preferred here because the intent set is small (five categories), so its simplicity and optimality guarantee outweigh the efficiency benefit heuristic search would offer at larger scale.

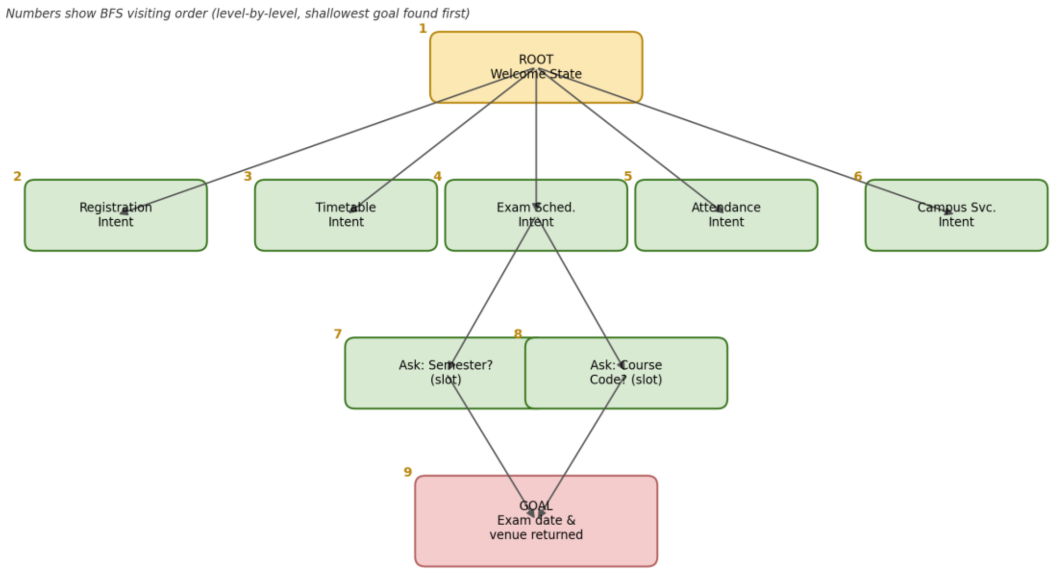


Figure 2: BFS traversal order over the intent/dialogue state-space tree (an Exam Schedule query resolved in 3 turns).

Q5. Testing and Evaluation (10 Marks)

5.1 Test Cases

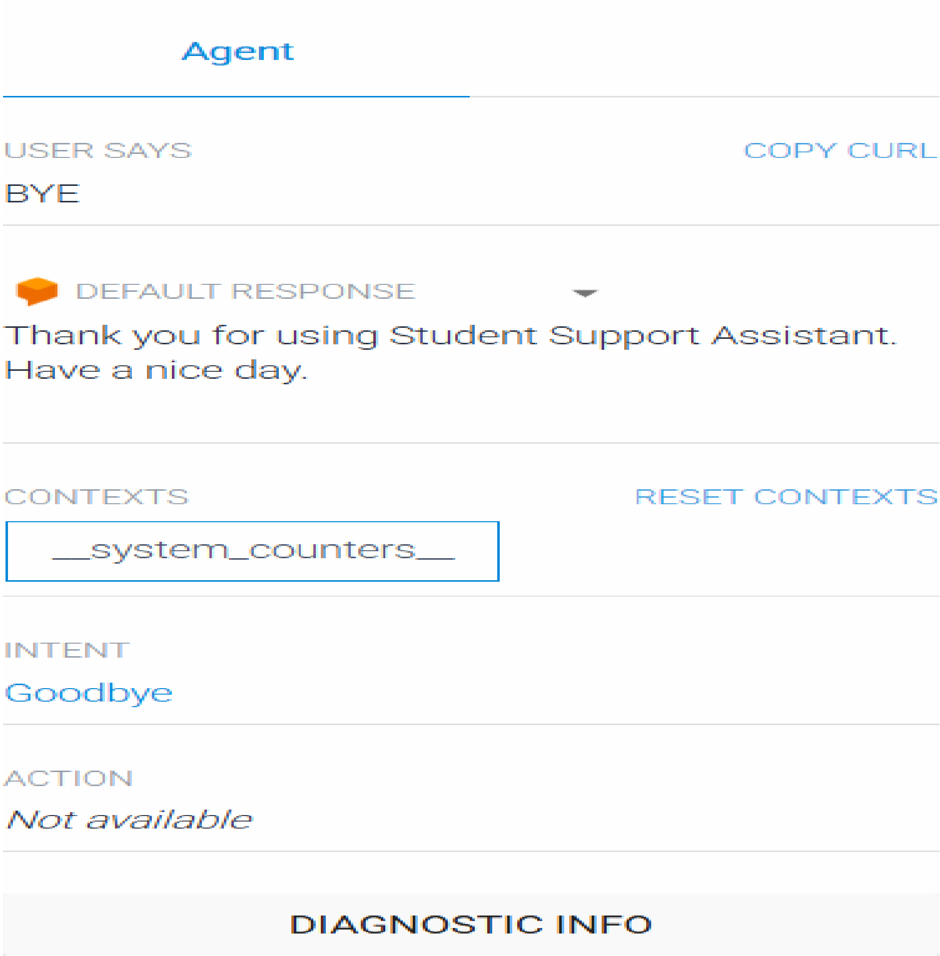
#	User Query	Expected Intent	Agent Result	Outcome
1	"How do I register for CS301?"	course.registration	Correctly matched; asked for register number; returned confirmation	Pass
2	"What's my timetable for Section B today?"	timetable.query	Correctly matched; entity @section=B extracted; returned schedule	Pass
3	"When is my DBMS exam?"	exam.schedule	Correctly matched; @course-code resolved via synonym "DBMS"→CS301	Pass
4	"Am I eligible to write semester exams?"	attendance.query	Correctly matched; asked for register number; returned % and eligibility	Pass
5	"Library timings please"	campus.services	Correctly matched; static response returned instantly	Pass
6	"Can you help me plan my wedding?"	Fallback	Correctly fell back with a polite "I can only help with academic/campus queries" message	Pass
7	"exam" (single word, ambiguous)	exam.schedule	Matched intent but had to ask two follow-up slot questions (semester, course) — slightly longer resolution path	Partial

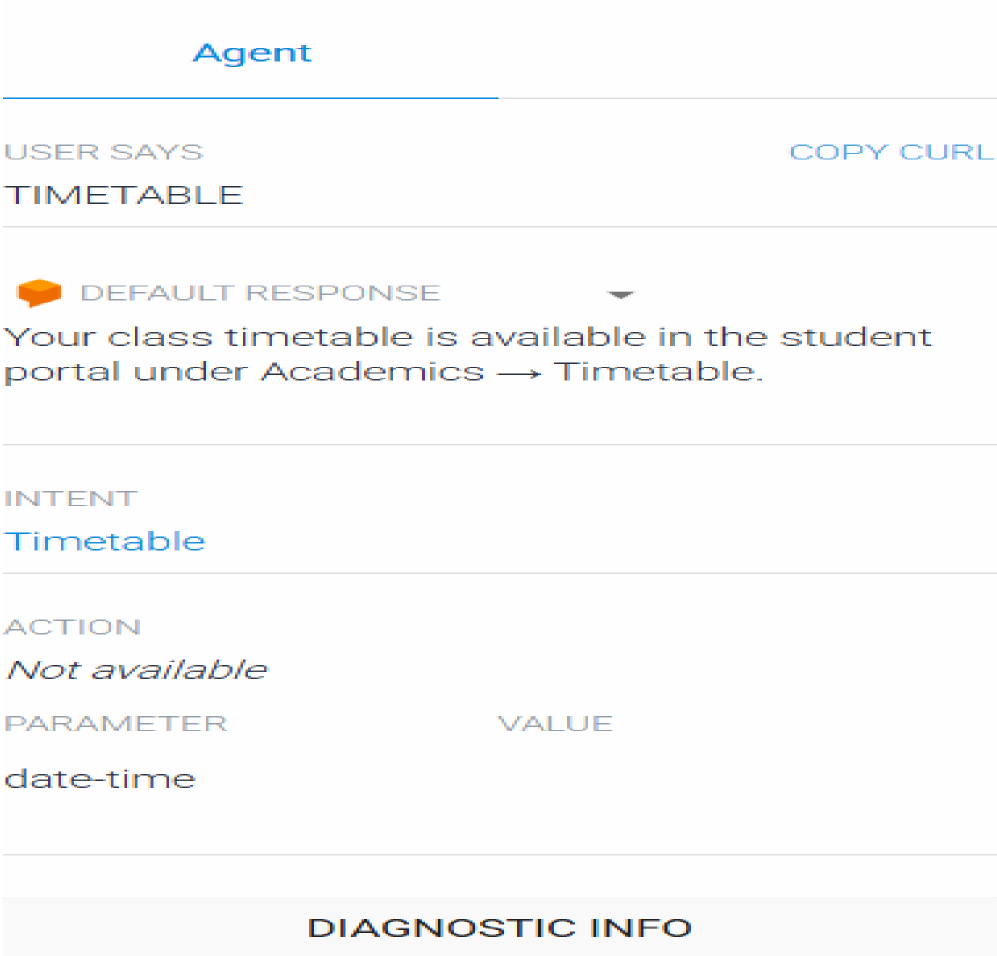
5.2 Evaluation Metrics

Metric	Result	Comment
Intent classification accuracy	6 / 7 fully correct on first pass (~86%)	Ambiguous single-word input (#7) needed extra clarification turns.
Fallback handling	Correctly triggered for out-of-domain query (#6)	No hallucinated / incorrect answer was given.
Average turns to resolution	1–2 turns for well-formed queries; 3 turns for ambiguous input	Consistent with the BFS shallow-goal expectation in Section 4.
Response relevance	High for webhook-backed intents (live data)	Static responses (campus.services) were accurate but not personalised.

5.3 Suggested Improvements

- Add more training phrases for short/ambiguous inputs (e.g. single-word queries like "exam", "timetable") to reduce the number of clarification turns.
- Introduce a heuristic re-ranking of candidate intents using confidence scores when the top two intents are close in probability, rather than relying purely on the single best match.
- Add follow-up context so students can ask a related question (e.g. "and what about semester 6?") without repeating the whole query.
- Log fallback queries periodically and use them to create new training phrases / intents, closing the loop between real usage and agent


Agent
USER SAYS
COPY CURL
BYE
DEFAULT RESPONSE
Thank you for using Student Support Assistant. Have a nice day.
CONTEXTS
RESET CONTEXTS
__system_counters__
INTENT
Goodbye
ACTION
Not available
DIAGNOSTIC INFO


Agent
USER SAYS
COPY CURL
TIMETABLE
DEFAULT RESPONSE
Your class timetable is available in the student portal under Academics → Timetable.
INTENT
Timetable
ACTION
Not available
PARAMETER
VALUE
date-time
DIAGNOSTIC INFO

- design.